CLINICAL TRIAL FAILURES · DISCONTINUATION RATE · ONCOLOGY PHASE 2,3 · STARTS 2015–2024

VEGF / VEGFR + PD-(L)1: 3 molecules stopped early

5 stopped trials across 3 molecules: Bevacizumab, Lenvatinib and Ningetinib. Against 5.8% for PD-(L)1 combinations without VEGF / VEGFR and 4.5% across all oncology Phase 2,3 trials in the same window. The rate counts trials that stopped early for efficacy, safety or benefit–risk reasons. Trials that ran to completion and missed their primary endpoint are counted separately below, never in the rate.

1.8%
5 of 278 closed trials (95% CI 0.8%–4.1%)

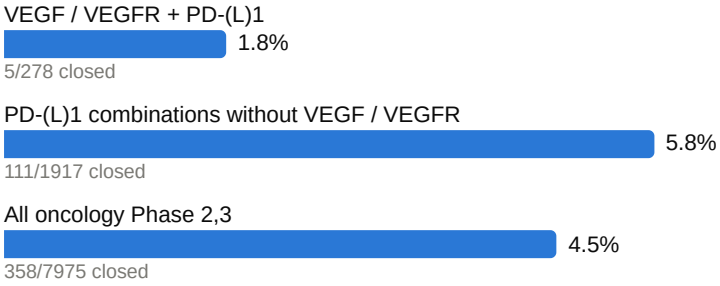
4 / 1 / 0
efficacy / safety / both (adds to 5)

25.8%
of 1076 trials have closed · 798 still open or unresolved

0.2%
stopped for a biological reason within 3 years of starting (95% CI 0.0%–1.1%; 401 still at risk; 1.5% area-wide)

9 of 18 **Completed and missed the primary endpoint.** Of 210 completed trials in this segment, 18 posted a primary result on ClinicalTrials.gov that can be read — the sponsor's own comparison held to the sponsor's own threshold, or its statement that the endpoint was missed: 9 missed and 5 met, 4 split across co-primary endpoints. Counted separately and never in the rate above; a floor, not a rate. 91 of the 186 completed trials that finished more than a year ago have posted no results at all — a gap in what can be known, not a sign of failure.

How this compares



What is in this class: 361 distinct experimental drugs across 1076 trials (antibody 1012, small molecule 891, unknown 191, protein 52). Most tested: BEVACIZUMAB (268), LENVATINIB (198), PEMBROLIZUMAB (160), RIVOCERANIB (140), CAMRELIZUMAB (130). A class groups drugs by what they act on; check that the grouping is one you would make before reading the rate as a property of the mechanism.

How much does this rest on one decision? The 5 stops came from 5 sponsor-asset programmes across 4 sponsors; the largest (Yale University) contributed 1; removing that programme's trials from both sides leaves 1.4%. Screened alongside 85 other segments, the rate is within what screening that many classes could produce (q=1); the molecules and trials that failed here are the record either way. A further 18 closed trials here were terminated with no cause recorded in the registry; if every one of them were biological the rate would be 8.3%. Read it as a screen worth checking against the underlying trials.

Bars show the share of closed trials stopped for biological reasons. Confidence intervals overlap where sample sizes are small — read the counts, not just the bars.

By start cohort

Trials started	Stops	Closed	Rate	95% CI
2015–2017	1	40	2.5%	0.4%–12.9%
2018–2020	2	131	1.5%	0.4%–5.4%
2021–2024	2	107	1.9%	0.5%–6.6%

Where the stops sit

Efficacy / futility: 4, Safety: 1.
Sponsors with most stops: Jiangsu Alphamab Biopharmaceuticals Co., Ltd (2), Yale University (1), Nasser Hanna (1), Vall d'Hebron Institute of Oncology (1).

The stopped trials

Trial	Ph	Sponsor	Experimental drugs	Stopped	Type	Registry stop reason
NCT04955743	2	Yale University	PEMBROLIZUMAB, LENVATINIB	2025-05	Efficacy	Lack of activity in the drugs

Trial	Ph	Sponsor	Experimental drugs	Stopped	Type	Registry stop reason
NCT05001724	3	Jiangsu Alphamab Biopharmaceuticals Co., Ltd	ERFONRILIMAB, LENVATINIB	2023-02	Efficacy	Because the efficacy data did not meet expectations, the sponsor decided to terminate the study.
NCT03838848	2	Jiangsu Alphamab Biopharmaceuticals Co., Ltd	ERFONRILIMAB, CARBOPLATIN, PEMETREXED, NINGETINIB	2022-07	Efficacy	Cohort A,B,C end enrollments. Cohort D and E were considered to have no significant clinical benefit at the SMC meeting and decided to terminate enrollment.
NCT03713944	2	Nasser Hanna	CARBOPLATIN, PEMETREXED, ATEZOLIZUMAB, BEVACIZUMAB	2021-12	Safety	Extreme toxicity, thromboembolic events
NCT02982694	2	Vall d'Hebron Institute of Oncology	ATEZOLIZUMAB, BEVACIZUMAB	2020-11	Efficacy	The results of the interim analysis showed that the efficacy in the MSS subgroup (MSI like) is inferior to the expected, and this situation is very unlikely to be reversed if the trial is fully completed.

Method

Denominator: ClinicalTrials.gov interventional Phase 2,3 oncology trials started 2015–2024 that have closed (completed or terminated). Numerator: terminated trials whose registry stop reason is classified as biological (efficacy, safety or benefit–risk) by Classification V2 — held-out precision 95.5%, recall 95.3% (n=600). Drugs are linked to ChEMBL and the NCI Thesaurus; 77% of industry oncology trials in this window carry a resolved drug target, and a trial without one cannot enter a mechanism class. Intervals are Wilson 95%.

Limits

Not a failure rate: completed trials that missed their endpoints are not in the rate. They are counted separately where the sponsor posted a primary result that can be read — most did not, so that count is a floor — and programmes discontinued after a completed trial do not appear. Stop reasons are sponsor-reported. This is a closed-trial proportion, not a time-to-event analysis: only 25.8% of trials in this segment have closed, and a trial that stops early enters the denominator sooner than one that runs to completion, which can inflate the rate in immature segments. Recent cohorts have fewer closed trials, so their rates are less stable. Research signals, not clinical or investment advice.

The whole cohort, trial by trial. The diligence report names every trial behind these figures — the ones still running and the ones that completed and missed their primary endpoint included — and places your molecule against each one that failed. Free during the beta: enter your work email at **clinicaltrialfailures.com/asset-check** and it opens straight away.

Sources: ClinicalTrials.gov (NLM); ChEMBL (EMBL-EBI, CC BY-SA 3.0); NCI Thesaurus (NCI); RxNorm/RxClass (NLM); SEC EDGAR. Classification, linkage and rates by Clinical Trial Failures. Rebuilt weekly; this brief covers trials started 2015–2024.